

ASSIGNMENT 9

Objective: Build a **Student Management Application** using JSP and JDBC that allows you to **add, Display, Update, delete, and View Details** of students stored in a relational database (e.g., MySQL).

1. Database Setup

- Create a database named college_db.
- Create a table student with the following fields:
 - id (INT, Primary Key, AUTO_INCREMENT)
 - name (VARCHAR(50))
 - email (VARCHAR(50))
 - course (VARCHAR(30))
 - age (INT)
 - cgpa (FLOAT)

2. Navigation Bar

- Every page should have (include) a **navbar** with only two options:
 - **Add Student** → link to the page where a new student can be added.
 - **Display Students** → link to the page that shows all students in a table.

3. Add Student

- Design a JSP page addStudent.jsp with a form to insert new student details (name, email, course, age). On submission, the data should be stored in the student table using JDBC.

4. Display Students (Read)

- Create a JSP page displayStudents.jsp that:
 - Fetches all student records from the database using JDBC.
 - Displays them in an **HTML table** with columns: ID, Name, Email, Course, Age, and **Actions**.
- In the **Actions column**, include three buttons/links for each student:
 - **Details** → Opens studentDetails.jsp to show full details of the selected student.
 - **Update** → Opens updateStudent.jsp with the student's current data pre-filled for editing. On submit, update the record in the DB.
 - **Delete** → Deletes the student record from the DB and reloads the display page.

addStudent.jsp :

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<%@ page import="java.sql.*" %>
<%@ include file="navbar.jsp" %>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
```

```

</head>
<body>
  <h2>Add New Student</h2>
  <form method="post">
    Name: <input type="text" name="name"><br>
    Email: <input type="email" name="email"><br>
    Course: <input type="text" name="course"><br>
    Age: <input type="number" name="age"><br>
    CGPA: <input type="text" name="cgpa"><br>
    <input type="submit" value="Save Student">
  </form>
  <%
    if(request.getMethod().equalsIgnoreCase("POST")){
      try {
        String driver = "org.postgresql.Driver";
        String url = "jdbc:postgresql://192.168.1.17/cse_db24";
        String username = "24BCSE57";
        String password = "24BCSE57";
        Class.forName(driver);
        Connection con = DriverManager.getConnection(url, username,
password);

        if(con!=null)
          System.out.println("Connection established successfully..");

        PreparedStatement ps = con.prepareStatement("INSERT INTO student01(name,
email, course, age, cgpa) VALUES(?,?,?,?,?)");
        ps.setString(1, request.getParameter("name"));
        ps.setString(2, request.getParameter("email"));
        ps.setString(3, request.getParameter("course"));
        ps.setInt(4, Integer.parseInt(request.getParameter("age")));
        ps.setFloat(5, Float.parseFloat(request.getParameter("cgpa")));
        ps.executeUpdate();
        out.print("Student added successfully!");
      }
      catch(Exception e) {
        out.print(e);
      }
    }
  %>
</body>
</html>

```

Add New Student

Name:	a
Email:	a@gmail.com
Course:	cse
Age:	21
CGPA:	6.7
Save Student	

displayStudents.jsp :

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8"%>
```

```
<%@ page import="java.sql.*" %>
```

```
<%@ include file="navbar.jsp" %>
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head><title>Display Students</title></head>
```

```
<body>
```

```
<h2>Student List</h2>
```

```
<table border="1">
```

```
<tr><th>ID</th><th>Name</th><th>Email</th><th>Course</th><th>Age</th><th>Action</th>
```

```
</tr>
```

```
<%
```

```
try {
```

```
    Class.forName("org.postgresql.Driver");
```

```
    Connection con =
```

```
DriverManager.getConnection("jdbc:postgresql://192.168.1.17/cse_db24", "24BCSE57", "24BCSE57");
```

```
    Statement st = con.createStatement();
```

```
    ResultSet rs = st.executeQuery("SELECT * FROM student01");
```

```
    while(rs.next()){
```

```
%>
```

```
<tr>
```

```
<td><%= rs.getInt("id") %></td>
```

```
<td><%= rs.getString("name") %></td>
```

```
<td><%= rs.getString("email") %></td>
```

```
<td><%= rs.getString("course") %></td>
```

```
<td><%= rs.getInt("age") %></td>
```

```
<td>
```

```
<a href="studentDetails.jsp?id=<%= rs.getInt("id") %>">Details</a> |
```

```
<a href="updateStudent.jsp?id=<%= rs.getInt("id") %>">Update</a> |
```

```

        <a href="deleteStudent.jsp?id=<%= rs.getInt("id") %>" onclick="return
confirm('Delete this record?')">Delete</a>
    </td>
</tr>
<%
    }
    con.close();
} catch(Exception e) { out.print(e); }
%>
</table>
</body>
</html>

```

deleteStudent.jsp :

```

<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>
    <%@ page import="java.sql.*" %>
<%
    String id = request.getParameter("id");
    try {
        Class.forName("org.postgresql.Driver");
        Connection con =
DriverManager.getConnection("jdbc:postgresql://192.168.1.17/cse_db24", "24BCSE57",
"24BCSE57");
        PreparedStatement ps = con.prepareStatement("DELETE FROM student01 WHERE
id=?");
        ps.setInt(1, Integer.parseInt(id));
        ps.executeUpdate();
        con.close();
        response.sendRedirect("displayStudents.jsp");
    } catch(Exception e) { out.print(e); }
%>
</body>
</html>

```

navbar.jsp :

```

<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>

```

```

<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>
    <div style="padding:10px; margin-bottom:20px;">
        <a href="addStudent.jsp">Add Student</a> |
        <a href="displayStudents.jsp">Display Students</a>
    </div>
</body>
</html>

```

StudentDetails.jsp :

```

<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8"%>
<%@ page import="java.sql.*" %>
<%@ include file="navbar.jsp" %>
<!DOCTYPE html>
<html>
<head><title>Student Details</title></head>
<body>
    <h2>Student Full Details</h2>
    <%
        String id = request.getParameter("id");
        try {
            Class.forName("org.postgresql.Driver");
            Connection con = DriverManager.getConnection
("jdbc:postgresql://192.168.1.17/cse_db24", "24BCSE57", "24BCSE57");
            PreparedStatement ps = con.prepareStatement("SELECT * FROM student01 WHERE
id=?");
            ps.setInt(1, Integer.parseInt(id));
            ResultSet rs = ps.executeQuery();
            if(rs.next()){
    %>
                <p><strong>ID:</strong> <%= rs.getInt("id") %></p>
                <p><strong>Name:</strong> <%= rs.getString("name") %></p>
                <p><strong>Email:</strong> <%= rs.getString("email") %></p>
                <p><strong>Course:</strong> <%= rs.getString("course") %></p>
                <p><strong>Age:</strong> <%= rs.getInt("age") %></p>
                <p><strong>CGPA:</strong> <%= rs.getFloat("cgpa") %></p>
    %>
            }
        }
    %>

```

```

        con.close();
    } catch(Exception e) { out.print(e); }
%>
</body>
</html>

```

updateStudent.jsp :

```

<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8"%>
<%@ page import="java.sql.*" %>
<%@ include file="navbar.jsp" %>
<!DOCTYPE html>
<html>
<head><title>Update Student</title></head>
<body>
    <h2>Update Student</h2>
    <%
        String id = request.getParameter("id");
        if(request.getMethod().equalsIgnoreCase("POST")){
            try { Class.forName("org.postgresql.Driver");
                Connection con = DriverManager.getConnection
("jdbc:postgresql://192.168.1.17/cse_db24", "24BCSE57", "24BCSE57");
                PreparedStatement ps = con.prepareStatement("UPDATE student01 SET name=?,
email=?, course=?, age=?, cgpa=? WHERE id=?");
                ps.setString(1, request.getParameter("name"));
                ps.setString(2, request.getParameter("email"));
                ps.setString(3, request.getParameter("course"));
                ps.setInt(4, Integer.parseInt(request.getParameter("age")));
                ps.setFloat(5, Float.parseFloat(request.getParameter("cgpa")));
                ps.setInt(6, Integer.parseInt(id));
                ps.executeUpdate();
                response.sendRedirect("displayStudents.jsp");
            } catch(Exception e) { out.print(e); }
        }
        try {
            Class.forName("org.postgresql.Driver");
            Connection con =
DriverManager.getConnection("jdbc:postgresql://192.168.1.17/cse_db24", "24BCSE57",
"24BCSE57");
            PreparedStatement ps = con.prepareStatement("SELECT * FROM student01 WHERE
id=?");
            ps.setInt(1, Integer.parseInt(id));
            ResultSet rs = ps.executeQuery();

```

```
        if(rs.next()){
%>
<form method="post">
    Name: <input type="text" name="name" value="<%= rs.getString("name") %>"><br>
    Email: <input type="email" name="email" value="<%= rs.getString("email") %>"><br>
    Course: <input type="text" name="course" value="<%= rs.getString("course") %>"><br>
    Age: <input type="number" name="age" value="<%= rs.getInt("age") %>"><br>
    CGPA: <input type="text" name="cgpa" value="<%= rs.getFloat("cgpa") %>"><br>
    <input type="submit" value="Update Record">
</form>
<%
    }
    con.close();
} catch(Exception e) { out.print(e); }
%>
</body>
</html>
```